

SNAPDRAGON AI LAB BUILD & PRESENT CHALLENGE 2026

Sahaay

सहाय · assistance

Hands-free and eyes-free control of a Windows PC, *entirely on the Snapdragon Hexagon NPU.*

Head and voice replace the mouse and keyboard; a conversational narrator replaces the screen. Six AI models, no cloud, no data leaves the laptop.



TWO PEOPLE

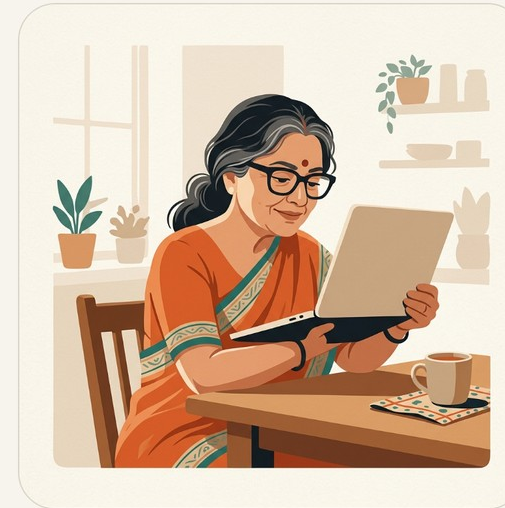
Two groups of people who cannot use a laptop the way it ships.



Arjun, 22

Engineering student · cerebral palsy

Cannot hold a trackpad steady. Every assignment, portal and chat needs **someone else's hands.**



Meena, 58

Shopkeeper · low vision, diabetic retinopathy

Can hear a screen reader list forty buttons. **Cannot see the photo her daughter sent,** or the chart in her bank app.

2.68 crore

Indians living with a disability
Census of India 2011

1 in 5

of them with a movement disability
Census of India 2011

1 in 5

of them with a visual disability
Census of India 2011

Composite personas drawn from published accessibility research; illustrations, not real people.

THE PROBLEM

Assistive technology today is expensive, depends on the cloud, and *stops at the basics*.

COST

₹1.5 lakh+

for an eye-gaze system

Plus a licence, a dongle, and often a subscription. Out of reach for a student on a ₹60,000 laptop budget, and rarely covered by insurance in India.

Tobii Dynavox, PCEye and comparable Indian resellers, 2026 list prices

PRIVACY

24 h/day

of face video and screen content

A cloud assistant has to stream the camera and the screen to a server. Face data is **sensitive biometric data under India's DPDP Act 2023**; the screen shows bank statements and medical records.

Digital Personal Data Protection Act 2023, s.2(t) and s.9; typical cloud speech and vision APIs

COVERAGE

0

built-in tools that describe a photo or the room

Screen readers read controls, not images. Voice control types and clicks, but still needs a hand for everything else. **Nothing ties head, voice, screen and camera together.**

Windows Narrator and Voice Access feature scope, Microsoft docs, 2026

THE IDEA

Run the entire assistive stack on the NPU that is *already inside the laptop.*

A Snapdragon Hexagon NPU runs face tracking, speech, a language model and a vision model at the same time, at a few watts, with nothing leaving the machine. HP ships it in the OmniBook that students already buy.

₹1.5 L+

eye-gaze hardware, licence,
cloud account

₹0

Sahaay: a download for the
laptop you already own



Two modes, one on-device brain: *hands-free control* and a *screen narrator*.

HANDS-FREE

Head moves the cursor. Voice types and commands.

For Arjun: motor impairment, injury, RSI

- **Nudge your head**, the cursor moves. Hold still, it stays. Calibrates itself.
- **Blink** to click, **hold still** to click, **open your mouth** to drag.
- **Dictate** into any app, in English or Hindi.
- **Say it:** *"open my downloads", "write a thank-you note to my teacher and save it."*
- **Or press a switch.** Any Arduino becomes a button interface.

NARRATOR

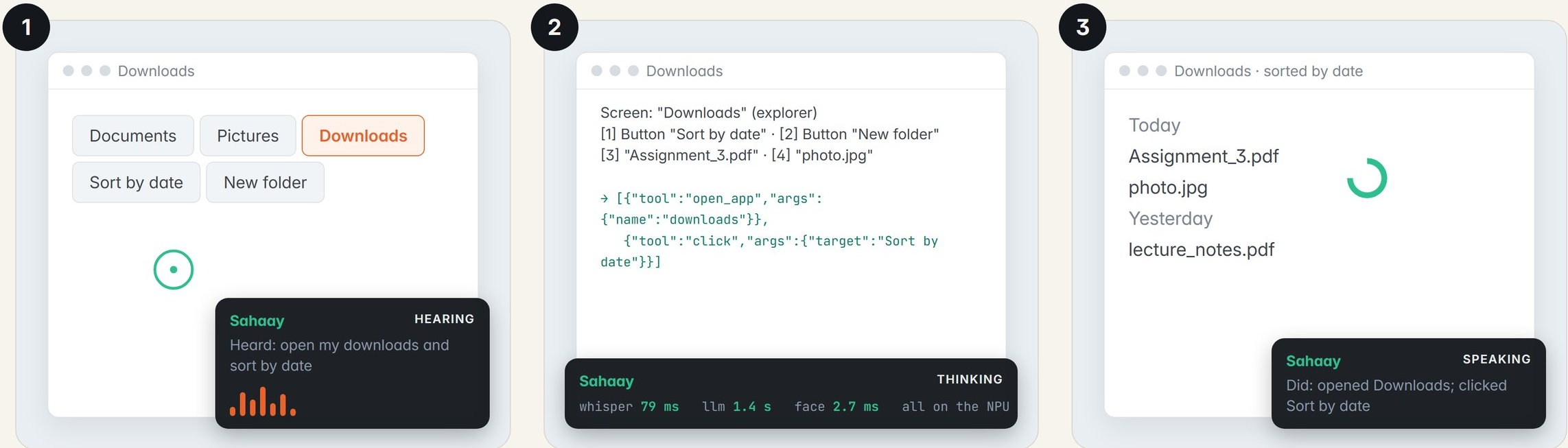
The screen and the camera, described in speech.

For Meena: low vision, blindness

- *"What's on my screen?"* Two spoken sentences: **which app, what content, what matters.**
- *"Read this."* The paragraph under focus, in her language.
- *"Describe the picture."* A vision model explains **photos, charts, canvases.**
- *"What do you see?"* The webcam view, labels and signs, **read aloud.**
- Every action is **confirmed by voice**, so she always knows what just happened.

ONE COMMAND, END TO END

How a spoken request becomes a *validated action* and a spoken confirmation.



ARJUN SAYS

"Open my downloads and sort by date."

ON THE NPU

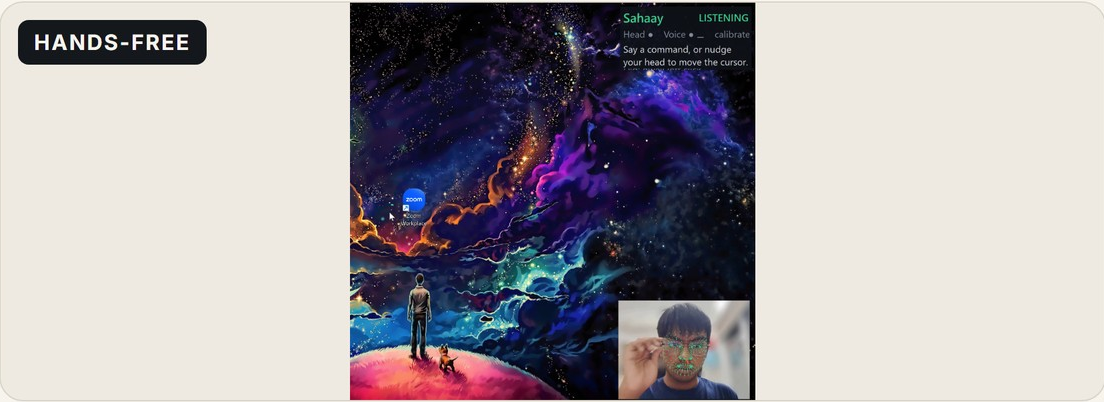
Speech becomes text, the screen becomes a list, the list plus the request becomes two safe tool calls.

SAHAAY SAYS

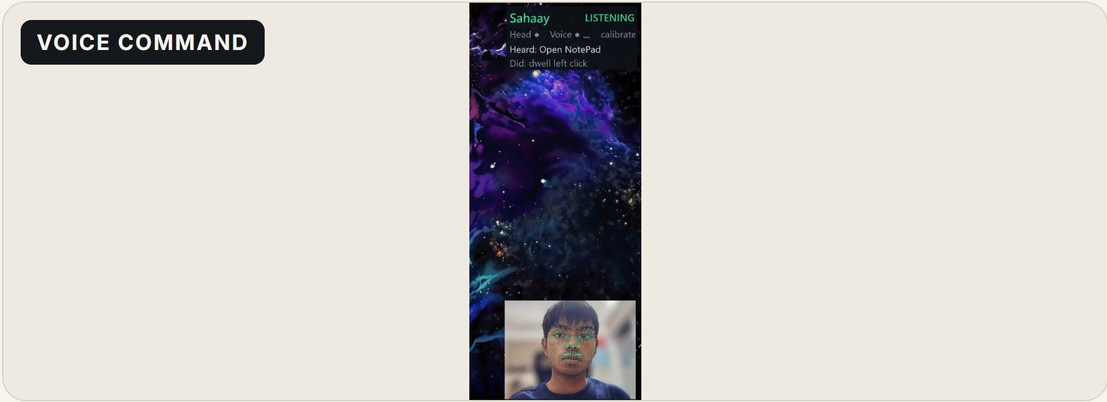
"Downloads is open, sorted by date. Newest first is Assignment 3."

THE PRODUCT, RUNNING

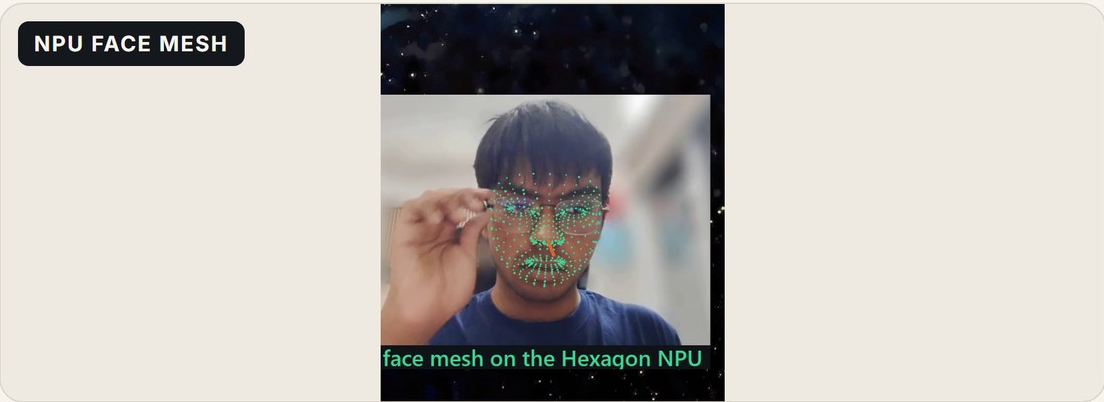
Screenshots from Sahaay on a Snapdragon X2 Elite laptop.



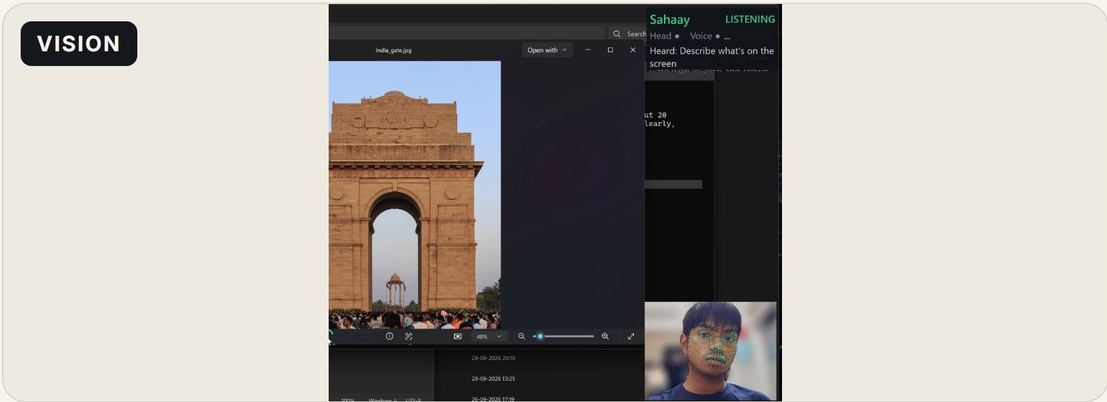
Head cursor with dwell ring and the live panel: face tracking on the NPU every frame.



"Open Notepad" heard by Whisper and executed; the panel shows what was heard and done.



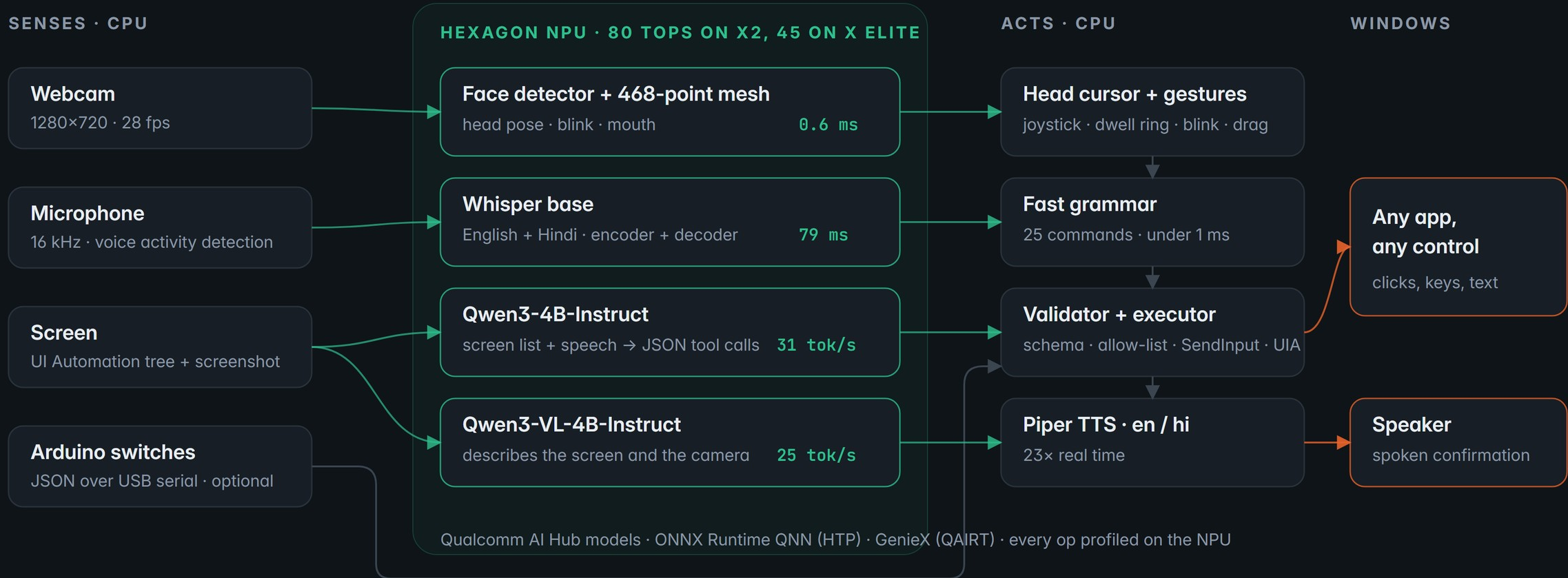
468 landmarks on the NPU, live in the corner window: head pose drives the cursor, blinks click.



"Describe what's on the screen": Qwen3-VL-4B on the NPU identifies India Gate and the crowd.

HOW IT WORKS

Six models on the Hexagon NPU. Capture, validation and input injection on the CPU. *The GPU untouched.*



— neural network, runs on the NPU — plain code, CPU — effect on Windows GPU: untouched, left to the user's apps

MEASURED PERFORMANCE

Every number is measured, and the NPU execution provider is *asserted* before it is reported.

Latency per stage, CPU vs NPU

Median of 30 runs on the development laptop (Snapdragon X2 Elite). Same graph, same inputs.

Face pipeline

detect, crop, 468-pt mesh, pose · live 720p frame



Face detector

BlazeFace 256×256



Face mesh

468 landmarks 192×192



Oryon CPU Hexagon NPU via QNN bars scaled per row

`tools/npu_check.py` asserts the active provider for every model. The usual `providers=[...]` call silently runs on the CPU with today's runtime; Sahaay selects the NPU device explicitly.

Sahaay

Speech and generative models, NPU only

Whisper ships as a precompiled QNN context; the language models run through GenieX on QAIRT.

79ms

to transcribe a 5.6 s command with Whisper base (21 ms encoder, 2.7 ms per token)

NPU

31tok/s

Qwen3-4B planning tool calls; 243 ms to first token with the screen in context

NPU

25tok/s

Qwen3-VL-4B describing the screen or the webcam; 269 ms to first token

NPU

10s

to load each generative model; face tracking is up in 2 s, so the cursor works almost immediately

NPU

Qualcomm AI Hub, reference devices, every op on the NPU

X2 Elite	face 0.4 ms · mesh 0.2 ms · Whisper enc 21.5 ms · dec 2.4 ms/token
X Elite (HP OmniBook)	face 0.7 ms · mesh 0.3 ms · Whisper enc 45.4 ms · dec 3.7 ms/token

09 · benchmarks/results.md and benchmarks/aihub_profiles.md in the repository

One command installs it on any Snapdragon X or X2 PC. *After that it never needs the internet.*

```
# any Snapdragon X / X2 PC · HP OmniBook, EliteBook · no admin rights
powershell -ExecutionPolicy Bypass -File install.ps1
# Python if missing · venv · models for your chipset, once · espeak-ng · Start Menu
shortcut
```



Private by construction

No account, no cloud, no telemetry. Camera frames, audio and screenshots **stay in RAM** and are discarded. A local log file for support, nothing else written.



Fits the OmniBook a judge owns

Hands-free mode within 16 GB. Both generative models resident on 32 GB (5.4 + 6.1 GB measured), or the 2B vision model on 16 GB. 45 TOPS is enough; 80 TOPS has headroom.



Usable without sight or hands

Global hotkeys, a tray icon, a heads-up panel that shows what was heard and done, **spoken confirmation of every action**, English and Hindi voices, and switch access through any Arduino, including the UNO Q from the AI Lab kit.

Where the cloud fails this user, and the *trade-offs* Sahaay accepts in return.

WHAT THE USER NEEDS	CLOUD ASSISTANT	SAHAAY ON THE HEXAGON NPU
Privacy of a face camera and the screen	Frames and screenshots leave the device continuously. Face data is sensitive biometric data under India's DPDP Act 2023 (s.2(t)); the screen carries bank and medical records	Processed in RAM, discarded. zero network calls after install; verifiable with Wi-Fi off
Cursor latency a person can tolerate	A network round trip before any inference. typically 50 to 150 ms to a cloud region from India, more on congested mobile links	2.7 ms per frame for the cursor; 79 ms for a spoken command. measured, benchmarks/results.md
Works in a hostel, a hospital ward, a train	Stops when the link drops. every command is a request; there is no offline mode	Fully offline. 6 GB of models cached once; nothing else fetched
Battery on a laptop that is the only interface	Video upload keeps the radio busy; inference still runs somewhere. continuous 720p upload is 1 to 3 Mbps over Wi-Fi or 4G	The CPU path would cost 25 ms per frame; the NPU does it in 2.7 ms and the CPU stays idle for the user's apps. measured, same frame
Cost to a student	Per-minute billing for speech, plus vision and language calls. Whisper API at \$0.006/min: three hours a day is about ₹2,700 a month, speech alone (₹84/\$)	Zero after the laptop. no subscription, no API keys

TRADE-OFFS WE ACCEPT **4-billion-parameter models** reason less well than frontier cloud models, so common commands bypass them through a fixed grammar and every model action is validated. **Each generative model takes 10 s to load** and 5 to 6 GB of RAM. **Whisper base mishears in noisy rooms**; fragments are dropped rather than acted on.

Seven problems between the models and a working product, *and their fixes.*

01 ONNX Runtime silently ran on the CPU while claiming the NPU

With the current plugin packaging, `providers=["QNNExecutionProvider"]` is accepted and ignored. **Fix:** register the QNN plugin, pick the NPU device through the device API, assert the active provider on every session, and ship `npu_check.py` so a reviewer can see it.

02 No OpenCV for Windows on ARM64

The camera and every image operation had to work without it. **Fix:** DirectShow through COM for capture (with `CoInitialize` on every worker thread), Pillow and numpy for the MediaPipe letterbox, affine crop and landmark mapping.

03 Whisper ships as a precompiled QNN context with no Python runner

Qualcomm's reference decode loop needs torch. **Fix:** re-implemented the encoder call and the 200-step KV-cache decoder loop in numpy: six layers of self-attention cache, causal mask and forced language prefix.

04 Piper's phonemiser is not built for ARM64

The wheel's espeak bridge fails to import. **Fix:** phonemise through the espeak-ng CLI and drive the VITS ONNX graph directly; English and Hindi voices at 23× real time.

05 Two GenieX models cannot initialise at the same time

Concurrent loads fail to create HTP contexts. **Fix:** a single loader thread brings up the language model, then the vision model, while the cursor and speech are already live.

06 A mis-heard fragment launched Windows Search

"Open it." from clipped audio became a search for "it". **Fix:** pronoun and length guards in the grammar, an allow-list of tools and elements, and no search fallback for short tokens.

07 QNN context caching and chipset generations

A cached compiled graph must be reused, not regenerated, and the development laptop's Hexagon v81 binaries do not run on the OmniBook's v73. **Fix:** reuse the cache when present, invalidate on mismatch, and ship separate model packs per chipset, both profiled on Qualcomm AI Hub.

ROADMAP

What ships next, in order.

- **Gaze on the NPU.** The AI Hub eyegaze model for coarse screen-region selection alongside head pose, for people with very limited neck movement.
- **Piper on the NPU, Hindi and Hinglish commands.** The AI Hub Piper recipe, and a command grammar that understands "neechе scroll karo".
- **Grounding from the vision model.** "Click the blue button" when an app has no accessibility tree: games, canvases, remote desktops.
- **Trials with an assistive-technology centre.** Calibration profiles for tremor and limited range.
- **MSIX package** for the Microsoft Store, and OmniBook presence-sensor auto-pause.

A ₹60,000 laptop that *anyone* can use, because the NPU is already inside it.

Everything in this deck is in the repository: code, benchmarks, installer, model packs and the reviewer tools that prove the NPU is doing the work.

github.com/Vision-jarvis/sahaay

Ruhan Srivastava

IIT Kharagpur · built on a Snapdragon X2 Elite with Qualcomm AI Hub, ONNX Runtime QNN and GenieX

